

Module 3: Strategic Equilibrium: Foundations & Applications

ECO364-THINKING STRATEGICALLY

Part III: Nash Equilibrium

MIXED STRATEGIES

Matching Pennies

		Player 2	
		Heads	Tails
Player 1	Heads	1, -1	-1, 1
	Tails	-1, 1	1, -1

- ▶ You and a friend simultaneously reveal a penny.
- ▶ If both pennies show heads or both show tails, player 2 pays player 1 \$1.
- ▶ If one penny shows heads and the other shows tails, player 1 pays player 2 \$1.
- ▶ Let's represent this with a payoff matrix.

Matching Pennies

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Matching Pennies

		Player 2	
		Heads	Tails
Player 1	Heads	★ 1, -1 → ★	★ -1, 1 → ★
	Tails	★ -1, 1 → ★	★ 1, -1 → ★

- ▶ Let's try to see if we have a (pure strategy) Nash Equilibrium.
- ▶ Let's see what are the best responses of the players.
 - ▶ If player 2 plays Heads, then player 1 should play Heads.
 - ▶ If player 2 plays Tails, then player 1 should play Tails.
 - ▶ If player 1 plays heads then player 2 should play Tails
 - ▶ If player 1 plays tails then player 2 should play heads
- ▶ As you can see, there is no correspondence of best responses.
- ▶ If one player plays a certain strategy, then there is always an optimal deviation to another (pure) strategy.

Matching Pennies

Player 1

		Player 2	
		Heads	Tails
Heads	★	1, -1	-1, 1
Tails	-1, 1	★	1, -1

- ▶ Games with a finite number of players and a finite number of actions always possess a Nash Equilibrium.
- ▶ Is something failing here?
- ▶ To find a Nash Equilibrium, we need to extend our concept of a strategy.
- ▶ **Definition:** A mixed strategy, is an assignment of a probability to each pure strategy.
- ▶ Any valid probability distribution (that adds up to one and has positive numbers) over a player's pure strategy is a mixed strategy.
- ▶ For example, for player 1, play Heads with probability 0.3 and Tails with probability 0.7 is a mixed strategy.
- ▶ For player 1, play heads with probability 1 and Tails with probability 0 is also a mixed strategy, albeit a degenerate one (is the pure strategy "play heads").
- ▶ Think of a mixed strategy as players choosing randomly, according to the mixed strategy, among their pure strategies

Matching Pennies

Player 1

		Player 2	
		Heads 0.5	Tails 0.5
Heads 0.5	★	1, -1	-1, 1
Tails 0.5	-1, 1	★	1, -1

Diagram illustrating the Matching Pennies game. Player 1 chooses Heads or Tails (0.5 probability each). Player 2 chooses Heads or Tails (0.5 probability each). The payoffs are shown in the matrix. Red arrows indicate that Player 2's best response is to choose the opposite of Player 1's choice. Blue arrows indicate that Player 1's best response is to choose the opposite of Player 2's choice. Stars mark the cells where both players are indifferent (the equilibrium).

An NE of this game is for both players to flip a coin (mix each pure strategy with probability .5)

- ▶ To find an equilibrium in this game, we will use the concept of a **mixed strategy Nash equilibrium**.
- ▶ In a mixed strategy Nash equilibrium, each player chooses independently a mixed strategy (a probability distribution over pure strategies).
- ▶ In equilibrium, each player's probability distribution makes all others indifferent between their pure strategies.
- ▶ For the matching pennies example, I can provide the following intuition:
 - ▶ Suppose that you were playing the game against a mind reader.
 - ▶ You would be destined to lose by playing any of your pure strategies, Heads or Tails.
 - ▶ No matter what you do, the mind reader can read your mind and adapt to any pure strategy that you are considering playing.
 - ▶ So what can you do to avoid losing?
 - ▶ The answer is to flip a fair coin!
 - ▶ In that way, the best the mind reader could do would be to win half the time!
 - ▶ The mind reader can't predict the flip of the coin!
- ▶ Back to the game: if both players randomize by flipping a coin, then both players are indifferent between their pure strategies. That is: if player 2 knows that player 1 flips a coin, then player 2 expects to win half the time and lose half the time, *regardless of whether 2 chooses heads or tails*. Similarly for player 1.
- ▶ Then each player indeed is indifferent between his or her pure strategies!

Matching Pennies

Player 1

		Player 2	
		Heads q	Tails $1-q$
Player 1	Heads p	★ 1, -1	-1, 1 ★
	Tails $1-p$	-1, 1 ★	1, -1 ★

Diagram illustrating the Matching Pennies game. The game is represented by a 2x2 payoff matrix. Player 1 chooses between Heads (p) and Tails (1-p). Player 2 chooses between Heads (q) and Tails (1-q). The payoffs are (Player 1, Player 2). Blue arrows indicate Player 1's best response: up for Heads and down for Tails. Red arrows indicate Player 2's best response: right for Heads and left for Tails. Stars mark the four possible outcomes.

An NE of this game is the mixed strategy profile $((0.5, 0.5), (0.5, 0.5))$

- ▶ Flipping a coin (mix with probability 0.5) doesn't always work for games different than matching pennies.
- ▶ There is a procedure to calculate mixed strategies for any game. I will use matching pennies here as an example of the procedure.
- ▶ Let's first work with player 1:
 - ▶ Suppose that player 1 expects player 2 to mix according to: Play heads with probability q , and Tails with probability $1-q$.
 - ▶ At an equilibrium, this probabilities must make player 1 indifferent between playing heads or tails.
 - ▶ Expected payoff for Player 1 of playing heads: $q \cdot 1 + (-1) \cdot (1-q) = 2q - 1$
 - ▶ Expected payoff for Player 1 of playing tails: $(-1) \cdot q + 1 \cdot (1-q) = 1 - 2q$
 - ▶ For player 1 to be indifferent: $2q - 1 = 1 - 2q$ which implies $q = 1/2$.
 - ▶ Thus in an eq. player 2 would put $q = 1/2$.
- ▶ Now for player 2:
 - ▶ Suppose that player 2 expects player 1 to mix according to: Play heads with probability p , and Tails with probability $1-p$.
 - ▶ At an equilibrium, this probabilities must make player 2 indifferent between playing heads or tails.
 - ▶ Expected payoff for Player 2 of playing heads: $(-1) \cdot p + 1 \cdot (1-p) = 1 - 2p$
 - ▶ Expected payoff for Player 2 of playing tails: $p \cdot 1 + (-1) \cdot (1-p) = 2p - 1$
 - ▶ For player 1 to be indifferent: $1 - 2p = 2p - 1$ which implies $p = 1/2$.
 - ▶ Thus in an eq. player 1 would put $p = 1/2$.

Example

- Consider the following strategic game:

		2	
		L	R
1	U	50, -10	30, 20
	D	20, 30	60, -20

- Find the set of Nash equilibrium points.
- Solution:
- First note that the game has no equilibrium in pure strategies. Let us use the equilibrium method to find the mixed strategy equilibrium. The surprising fact is that we equate the payoff of 2 when we compute the mixed strategy of 1. Namely, the equilibrium mixed strategy of 1 depends only on the payoff structure of 2 (and similarly, the equilibrium mixed strategy of 2 depends only on the payoff structure of 1).

Example

		2	
		L	R
1	p U	50, -10	30, 20
	1-p D	20, 30	60, -20

- ▶ Suppose that 1 chooses the pure strategy U with probability p and he chooses the pure strategy D with probability $1-p$
- ▶ If 2 chooses L she obtains an expected payoff:
 $-10 \cdot p + 30 \cdot (1-p) = 30 - 40p$
- ▶ If 2 chooses R she obtains an expected payoff:
 $20 \cdot p + (-20) \cdot (1-p) = (-20) + 40p$
- ▶ In equilibrium 2 must be indifferent between her two strategies L and R. Why? If L yields her a payoff higher than R she should choose L for sure (an average of two different numbers is smaller than the highest of the two)
- ▶ Consequently the two expected payoffs should be equal:

$$\begin{aligned} 30 - 40p &= -20 + 40p \\ 50 &= 80p \\ p &= 5/8 \end{aligned}$$
- ▶ Namely, 1 should play U with probability $5/8$ and D with probability $3/8$. In this case 2 obtains an expected payoff of 5, $30 - 40 \cdot (5/8) = 5$, whether she chooses L or R and she has no incentive to deviate.

Example

		2	
		q	1-q
1		L	R
	U	50, -10	30, 20
	D	20, 30	60, -20

- ▶ As for the mixed strategy equilibrium $(q, 1-q)$ of 2 we equate the expected payoff of 1 obtained by his two pure strategies U and D
- ▶ U : $50q + 30(1-q) = 30 + 20q$
- ▶ D : $20q + 60(1-q) = 60 - 40q$
- ▶ $30 + 20q = 60 - 40q \rightarrow 60q = 30 \rightarrow q = 1/2$
- ▶ That is, 2 should equally randomize between L and R. The payoff of 1 is $30 + 20 \cdot (1/2) = 40$
- ▶ The unique Nash equilibrium of this game is thus
- ▶ $((5/8, 3/8), (1/2, 1/2)) \rightarrow$ the payoffs from this NE are $(40, 5)$

Example: The hidden pearl

- ▶ There are two large drawers, filled with clutter of miscellaneous objects. Player 1 hides a pearl in one of them. Player 2, a burglar, then has one minute to open a drawer and look for the pearl. If he opens Drawer A and the pearl is there he finds it with probability $1/2$. If he opens Drawer B and the pearl is there, he gets it with probability $1/3$. If he opens the wrong drawer he gets nothing of value. Finding the pearl is worth \$50 to the burglar and (-\$50) to 1. Failing to find the pearl is worth (-\$5) to the burglar. Find the Nash equilibrium of this game. Assume that 2 has no time to open more than one drawer.
- ▶ Solution:
 - ▶ Player 1 has two strategies: “Hide in A” or “Hide in B”.
 - ▶ Player 2 has also two strategies “Open A” and “Open B”.
 - ▶ Let us find the expected payoffs as a function of their decisions.
 1. Suppose both players choose A.
 - ▶ Then the expected payoff of 1 is:
 - ▶ $(1/2)*(-50) + (1/2)*0 = -25$
 - ▶ The expected payoff of 2 is:
 - ▶ $(1/2)*(50) + (1/2)*(-5) = 22.5$
 - 2. Suppose both players choose B.
 - ▶ Then 1 obtains
 - ▶ $(1/3)*(-50) + (2/3)*0 = (-50/3)$
 - ▶ Player 2 obtains
 - ▶ $(1/3)*(50) + (2/3)*(-5) = (40/3)$

Example: The hidden pearl

- We can summarize the previous slide in the following table

		2	
		q	1-q
1	A	A	B
	B	A	B
p		-25, 22.5	0, -5
1-p		0, -5	$-(50/3), (40/3)$

- Observe that there are no pure strategy equilibrium. We use the equalization method to find the mixed strategy equilibrium $(p, 1-p)$ and $(q, 1-q)$ of 1 and 2 respectively
- $22.5 \cdot p + (-5) \cdot (1-p) = -5p + (40/3) \cdot (1-p)$
- Multiply by 6 the whole equation above to get: $135p - 30 \cdot (1-p) = -30p + 30(1-p)$
- $p = 2/5$
- Hence 1 hides the pearl in A with probability $2/5$ and hides it in B with probability $3/5$. As for 2
- $-25q + 0 \cdot (1-q) = 0q + -(50/3) \cdot (1-q)$
- Multiply by 3 the whole equation above to get: $-75q = -50 + 50q$
- $q = 2/5$
- That is, 2 searches the pearl in A with probability $2/5$.
- The expected payoff of 1 is $-25 \cdot (2/5) = -10$ and the expected payoff of 2 is $-30 \cdot (2/5) + 80 \cdot [1 - (2/5)] = 36$. The only equilibrium is $((2/5, 3/5), (2/5, 3/5)) \rightarrow$ with payoffs $(-10, 36)$

Application: Penalty kicks in soccer

- ▶ There have been 22 penalty kick shootouts in World Cup history. Six have occurred in the second round, 10 in the quarterfinal, four in the semifinal (including both semifinals in the 1990 World Cup, considered by many to be the worst in the tournament's history) and two have decided the title. The first final to be decided via shootout was in 1994, when Brazil topped Italy thanks to Italian [Roberto Baggio's infamous miss.](#)
- ▶ The other final decided by shootout was in 2006, when Italy topped France 5-3 after a game in which French captain Zinedine Zidane head-butted Italian Marco Materazzi, [earning a red card](#) and rendering himself unavailable for the shootout. Fabio [Grosso wound up sinking the winner for Italy.](#)
- ▶ We are going to look at some numbers that are approximately the probabilities of scoring when you kick the penalty kick in different directions.

Application: Penalty kicks in soccer

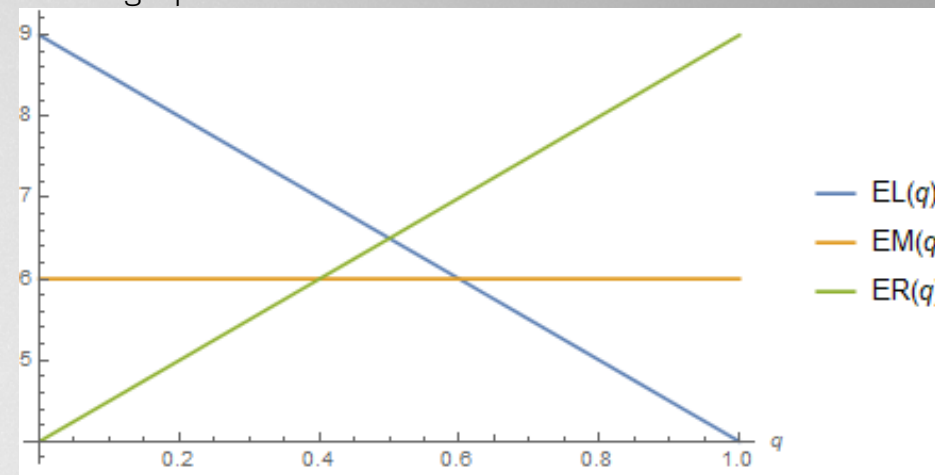
- ▶ There are three ways, the goal—the attacker could kick the ball. He could kick the ball to the left, the middle, or the right.
- ▶ The goalie can dive to the left or the right.
- ▶ Consider the following table:

		Goalie	
		l	r
Shooter	L	4, -4	9, -9
	M	6, -6	6, -6
	R	9, -9	4, -4

- ▶ The idea here is that the number 4 represents 40% chance of scoring if you shoot the ball to the left and the goal keeper dives to the left.
- ▶ The payoff for the shooter is his probability of scoring and the payoff for the goal keeper is the negative of that.
- ▶ It is clear from this matrix that no player has a strictly dominated pure strategy.
- ▶ But it turns out here that for the shooter, the strategy M is strictly dominated by a *mixed strategy*.
- ▶ Notice that if the shooter puts probability $p=0.5$ to L and probability $1-p=0.5$ to R, and probability 0 to M, then irrespective of what the goalie is doing, the expected payoff of this strategy is always greater than the payoff of playing pure strategy M.
- ▶ Expected payoff of $(0.5, 0, 0.5)$ against l: $4 \cdot 0.5 + 9 \cdot 0.5 = 6.5$
- ▶ Expected payoff of $(0.5, 0, 0.5)$ against r: $9 \cdot 0.5 + 4 \cdot 0.5 = 6.5$
- ▶ Both expected payoffs are greater than the payoff of playing M, irrespective of what the Goalie is doing!
- ▶ Therefore, M is dominated by the mixed strategy $(0.5, 0, 0.5)$ of player 1!

Application: Penalty kicks in soccer

- ▶ It turns out that sometimes, a strategy which is not strictly dominated by any pure strategy, can be strictly dominated by a mixed one.
- ▶ Recall: *if a strategy is strictly dominated, then it is never a best response to any belief.*
- ▶ Suppose the shooter believes that the goalie will play l with probability q and r with probability $1-q$.
- ▶ We can represent this in a graph. For any q in the interval $[0,1]$, we will calculate the expected payoff of the shooter for each strategy as a function of his beliefs q .
- ▶ Here is the graph:



		Goalie	
		q l	$1-q$ r
Shooter	L	4, -4	9, -9
	M	6, -6	6, -6
	R	9, -9	4, -4

- ▶ $EL(q) = 4 \cdot q + (1-q) \cdot 9 = 9 - 5q$
- ▶ $EM(q) = 6 \cdot q + (1-q) \cdot 6 = 6$
- ▶ $ER(q) = 9 \cdot q + (1-q) \cdot 4 = 4 + 5q$
- ▶ Notice that no matter what the belief of (q) is of the shooter, playing Middle is **never a best response**.
- ▶ In particular for $q < 0.5$ the best response to any such belief is to play Left. On the other hand, for any belief such that $q \geq 0.5$, playing Right is the best response. M never is for any belief.
- ▶ Thus, M should never be played in equilibrium!

Application: Penalty kicks in soccer

- ▶ To sum up: if a strategy is strictly dominated by other mixed or pure strategy, then it is never a best response to any belief.
- ▶ In the penalty kick game, we can eliminate M because it is never a best response.
- ▶ To calculate the equilibrium, we will use the equalization method.

		Goalie	
		q l	$1-q$ r
Shooter	p L	4, -4	9, -9
	$1-p$ R	9, -9	4, -4

- ▶ Equating the expected payoffs of the shooter if the goalie plays $(q, 1-q)$ gives :
- ▶ $9-5q=4+5q \rightarrow q=1/2$
- ▶ Similarly, for the goalie:
- ▶ $El(p)=4*p+(1-p)*9=9-5p$
- ▶ $Er(p)=9*p+(1-p)*4=4+5p$
- ▶ Equating them gives:
- ▶ $9-5p=4+5p \rightarrow p=1/2$
- ▶ So, a Nash equilibrium of this penalty kick game is $((0.5,0,0.5), (0.5,0.5))$. Remember that since M is dominated, we are putting probability 0 on it, hence $(0.5,0,0.5)$.